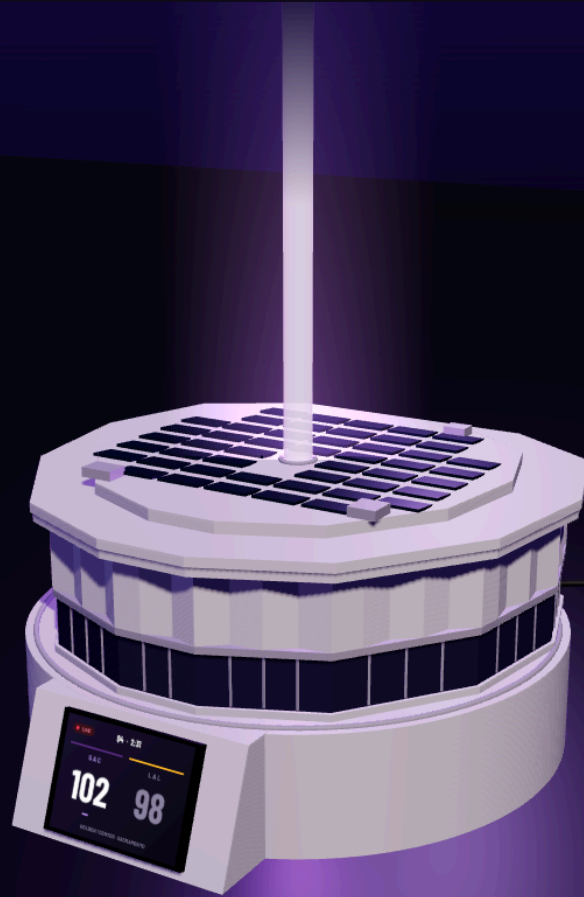




LIGHT THE BEAM

Build guide: printing, flashing, assembly and setup. Plan on two evenings: one for printing and flashing, one for assembly. No soldering.

READ EACH PART BEFORE STARTING IT

How this guide is organised

The build has five parts. Each part starts with what you need on the table and ends with a check so you know it worked before moving on. Times assume the prints are done.

Part 1 · Print the parts	p. 3–4 · ~18 h printer time	Part 2 · Flash the board	p. 5 · 20 min
Part 3 · Build the beam	p. 6–8 · 25 min	Part 4 · Arena and plinth	p. 9–12 · 30 min
Part 5 · Power on and set up	p. 13 · 10 min	Using it · Troubleshooting · Specs	p. 14–15

Parts list

Letters are used throughout the guide. Printed parts are A–G; bought parts are H–K (see [docs/bom.md](#) for links).

A**Arena base, 220 mm**silver PLA · [Printables file](#)**B****Arena lid**

silver PLA · with 12.4 mm beam hole

C**Plinth body**silver PLA · [plinth.stl](#)**D****Plinth bottom lid**silver PLA · [plinth_lid.stl](#)**E****Beam tube, 250 mm**white PLA · [beam_tube.stl](#)**F****Strip spine**white PLA · [beam_spine.stl](#)**G****Tube socket**white PLA · [beam_socket.stl](#)**H****Display board**ESP32-2432S028R + USB cable + 2
pigtail leads**I****LED strip**WS2812B 5 V, 60/m, 1 m · you use 15
LEDs**J****3 lever connectors**

WAGO 221-413

K**USB power adapter**

5 V, 2 A or more, USB-A

—**Optional**USB-C panel jack · 74AHCT125 level
shifter

Tools and supplies

- ✓ Scissors or flush cutters
- ✓ Ruler or calipers
- ✓ Masking tape and a pencil
- ✓ A computer with a USB port (for flashing, once)
- ✓ Hot glue gun (or super glue and patience)
- ✓ Small flat screwdriver (helps with the lever nuts)
- ✓ A phone and your home WiFi password
- ✓ Isopropyl alcohol wipe (for the socket glue joint)

Get the files and measure the arena

1 Download the arena from Printables

5 MIN

1. Open `printables.com/model/338758` and sign in (a free account is required to download).
2. Click **Download**, then in the file list take only the two files for the **220 mm** size: the **base** and the **lid with no hole**. Skip the 190 and 296 mm files and the pen-holder remix.
3. Save both into the project folder `models/printables/`.

Why the no-hole lid? The author's hole is sized for a store-bought light bar. Our beam tube is 12 mm, so you'll cut the hole yourself in step 3. If you'd rather not, download the holed lid too and measure it; if it's 12–13 mm it will work.

2 Measure the base for the plinth

10 MIN

The plinth is generated to fit your arena, so it needs two numbers from the base file.

1. Open **Bambu Studio**, drag the base STL onto the plate. Click the model and check the size readout in the right panel: you should see roughly $220 \times 200 \times 70$ mm.
2. Choose the **Measure** tool (ruler icon in the top toolbar). Click the flat back edge of the footprint, then the front prow: that's the **depth**. Then the widest left-right span: the **width**. Write both down.
3. Look at the back wall near the floor for the author's **wire hole**. Note which side it's on when the flat edge faces you: that is the back, angle 180° .
4. Open `models/plinth.scad` in a text editor. Set `arena_d` to the **larger** of your two numbers, and `arena_wire_a` to 180 (back) or wherever the hole is. Save.

CHECK Your `arena_d` should be within a few mm of 220. If it's wildly different you picked the wrong size file.

3 Make the plinth, tube, spine and socket files

5 MIN

1. Install **OpenSCAD** if you don't have it (free, opencscad.org, drag to Applications).
2. In Terminal: `cd models && make`. Six STLs appear in `models/exports/`. The plinth takes a minute to render; that's normal.

```
cd ~/dev/personal/prints/sacramento_kings_beam/models
make                # writes exports/plinth.stl, plinth_lid.stl, plinth_test.stl, beam_tube.stl,
                    beam_spine.stl, beam_socket.stl
```

Slice and print

4 Print the plinth test slice first

40 MIN PRINT

1. In Bambu Studio, import `exports/plinth_test.stl`. It's just the console from the front of the plinth. Print it in silver PLA, 0.2 mm, default walls, flat face down.
2. When it's done, slide the display board **H** into the pocket from the open bottom, screen toward the window. It should slide with light finger pressure and stop with the glass against the window lip.
3. Look at it from the front. The picture area of the screen (not the black border) should be centred in the opening with an even margin.

CHECK Board slides in, screen centred, no rattle. If it's too tight, raise `pocket_clear` in `plinth.scad` by 0.2. If the picture is off to one side, change `screen_ofs_x` by the amount it's off (in mm; positive moves the window right). Re-run `make` and reprint only the test slice.

5 Cut the beam hole in the lid

3 MIN

1. Import the **no-hole lid**. Right-click it → **Add Negative Part** → **Cylinder**.
2. In the negative part's settings set size **12.4 × 12.4 × 20 mm** and position it dead centre of the lid (X 0, Y 0 relative to the lid), sticking through the top and bottom.
3. Slice and confirm the preview shows a round hole through the lid.

12.4, not 12. The tube is 12 mm; the extra 0.4 is print tolerance so it slides without forcing.

6 Print everything

~18 H PRINTER TIME

PART	FILE	FILAMENT	ORIENTATION	SETTINGS	TIME
A · Arena base	Printables 220 base	silver	as downloaded, floor down	0.2 mm · 3 walls · 15 % · no supports	~9 h
B · Arena lid	Printables no-hole lid + your cut	silver	flat side down	0.15 mm for the roof detail	~2 h
C · Plinth body	<code>plinth.stl</code>	silver	upside down (flat top on the plate)	0.2 mm · 3 walls · 15 % gyroid	~5 h
D · Bottom lid	<code>plinth_lid.stl</code>	silver	flat side down	0.2 mm	~40 min
E · Beam tube	<code>beam_tube.stl</code>	white	foot down	Spiral vase on · 0.4 nozzle · 2 bottom layers · 0.2 mm	~1 h
F · Spine, G · Socket	<code>beam_spine.stl</code> , <code>beam_socket.stl</code>	white	spine flat, socket open side up	0.2 mm	~20 min

Vase mode for the tube only. In Bambu Studio: Process → Quality → **Spiral vase**. It prints one continuous wall, which is what makes the tube glow evenly. Don't enable it for anything else.

Plate layout. The base alone fills most of the plate. Print it on its own, then lid + spine + socket together, then the plinth on its own, then the tube on its own (a 250 mm vase print doesn't like neighbours).

Put the firmware on the display board

Do this before assembly, while the board is easy to plug in. You only ever do it once; WiFi is set up from your phone later.

7 Install PlatformIO (once)

5 MIN

1. Open Terminal and run the command below. It installs the build tool into your user folder; nothing system-wide changes.

```
python3 -m pip install --user platformio
export PATH="$HOME/Library/Python/3.9/bin:$PATH"    # add this line to ~/.zshrc so it sticks
```

8 Connect the board and flash the demo

5 MIN

1. Plug the board **H** into your Mac with its USB cable. Use the **micro-USB** port (some boards also have USB-C; either works for flashing).
2. In Terminal, go to the firmware folder and flash the **demo** build first. The first build downloads the ESP32 toolchain (a few hundred MB) and takes a few minutes.

```
cd ~/dev/personal/prints/sacramento_kings_beam/firmware
pio run -e cyd_demo -t upload
```

3. Watch the end of the output for **SUCCESS**. The board restarts by itself.

CHECK The screen lights up and cycles every 9 seconds: **NEXT GAME** (logos, countdown) → **LIVE** (big score) → **FINAL · LIGHT THE BEAM**. Tap the screen: nothing visible yet (no strip), but the touch is being read.

If the upload fails with "no serial port": unplug, plug back in, run again. If it still fails, hold the board's **BOOT** button while the upload starts, release after "Connecting...". If the picture is upside down or the colours are inverted, see Troubleshooting on page 15 and reflash.

9 Flash the real firmware

3 MIN

```
pio run -e cyd -t upload
```

CHECK The screen shows the Kings logo, "STARTING", then **WIFI SETUP** with four instructions. That's correct: it has no WiFi yet. Unplug the board; you'll set up WiFi in Part 5.

Want to watch it think? `pio device monitor -b 115200` shows the log: WiFi, each ESPN fetch, and the beam decisions.

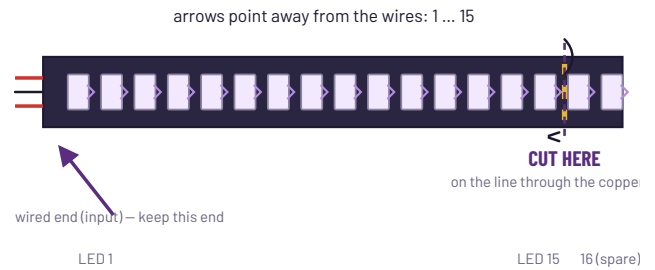
Cut and mount the strip

10 Cut the strip to 15 LEDs

5 MIN

1. Unroll the LED strip I. One end has three wires already attached (red, green, white or black) and usually a small white plug. **That end stays.**
2. Look closely at the strip: small arrows are printed along it. They point **away** from the wired end. Data travels in the arrow direction; the wired end is the input.
3. Starting at the wired end, count **15 LEDs**. Put a piece of masking tape after the 15th.
4. Between LED 15 and LED 16 there is a set of three copper pads with a line through the middle. Cut **exactly on that line** with scissors, straight across.
5. Put the rest of the roll away; it's spare.

Cut on the line, not through an LED. Cutting anywhere else kills the LED on each side of the cut. If you slip, cut on the next line and live with 14 LEDs; the firmware doesn't mind.



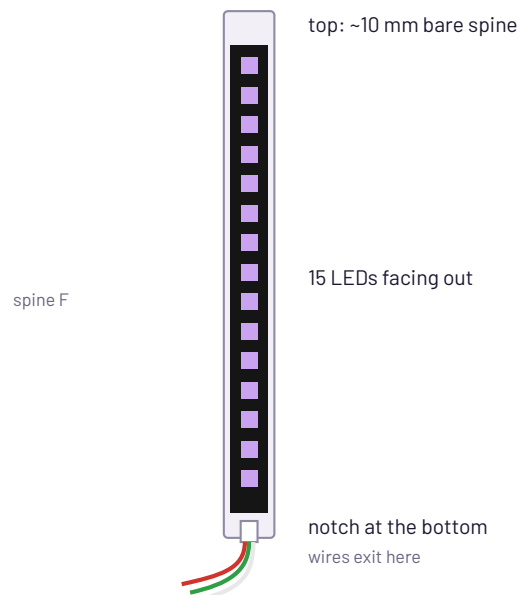
Count 15 LEDs from the wired end. Cut on the pad line after the 15th, never through an LED.

11 Stick the strip to the spine

5 MIN

1. Take spine F. It's a flat white bar with a small notch at one end. The notch is the **bottom**.
2. Wipe the spine with the alcohol wipe and let it dry; the strip's adhesive needs a clean surface.
3. Peel 3 cm of the strip's paper backing. Line the strip up on the flat face of the spine so the **wired end is at the notch** and the three wires exit through it.
4. Press down the first few LEDs, then peel and press a bit at a time along the length, keeping the strip straight and centred. It should end about 10 mm short of the top.
5. Rub the whole length down with a thumb.

CHECK Wires exit at the notched bottom, LEDs face away from the spine, strip runs straight to near the top.



Into the tube, and a first light

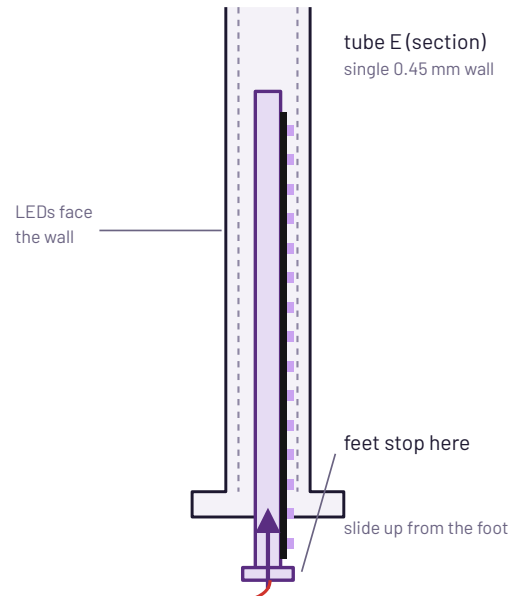
12 Slide the spine into the tube

3 MIN

1. Hold tube **E** with the wide foot flange **down**. The tube is open at the foot.
2. Feed the **top** of the spine (the end without wires) into the foot opening and push it up, LEDs facing the tube wall (any direction; it's round).
3. Push until the small feet at the bottom of the spine stop against the inside of the flange. The three wires should hang out of the foot.

Stuck halfway? The single-wall tube can be slightly oval. Rotate the spine 90° and try again, or gently squeeze the tube round. Never force it; the wall is 0.45 mm.

Optional: one small dab of hot glue where a spine foot meets the flange stops the spine sliding out when you tip the tube. Keep glue off the strip.



The spine goes in top-first through the foot opening; its feet stop against the inside of the flange.

13 Test the beam before it goes inside the arena

5 MIN

It's much easier to fix a wiring mistake now than after the glue.

1. Find the two small **pigtail leads** that came with board **H**: thin cables with a tiny white plug on one end and bare tinned wires on the other. Plug one into the 4-pin socket labelled **P1** (near the USB port) and one into **CN1** (the 4-pin socket on the edge).
2. Identify the wires using the printed labels next to each socket. You need three: **VIN** and **GND** from P1, and **I022** from CN1. Put a bit of tape on each and write the name on it. Fold any unused wires back and tape them so they can't touch anything.
3. Open the three lever connectors **J** (flip the orange levers up). Make three joints, one per connector, matching the diagrams on the next page: **VIN** ↔ **red**, **GND** ↔ **white/black**, **I022** ↔ **green**. Push each bare wire in until it stops, then close the lever. The board's pigtail wires are thin: fold the bare end back on itself once so it's doubled before inserting.
4. Tug each wire gently. None should pull out.
5. Plug the board into the USB adapter **K**. It boots to WIFI SETUP. **Tap the screen once.**

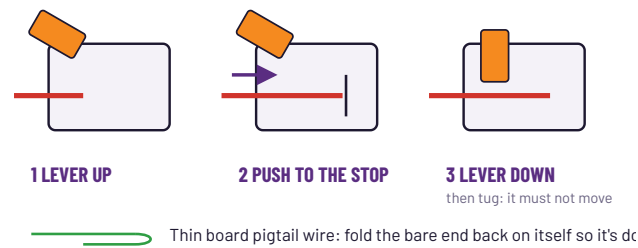
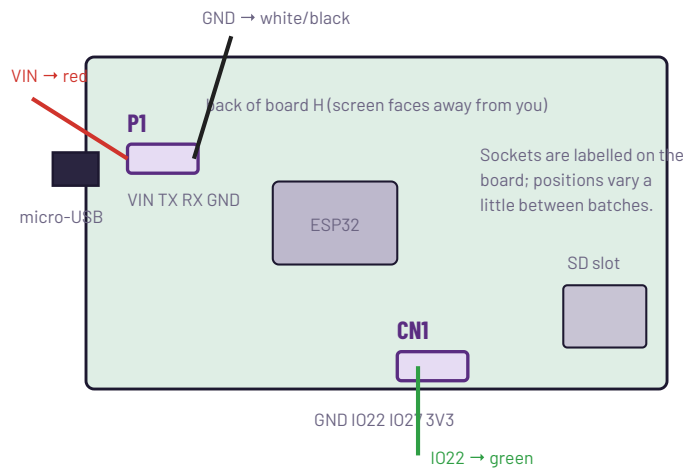
CHECK The strip lights purple from the bottom LED up, then all 15 hold purple with a slow shimmer. Tap again: they go dark. Hold your finger on the screen for a second: brightness steps down; hold twice more: back to full.

Nothing lights? Unplug first. Check green is on I022 (not I027) and red is on VIN (not 3V3). Check the strip's arrows point away from the wired end. **Only the first LED lights, or colours flicker?** The data signal is marginal: shorten the green lead, or add the optional 74AHCT125 level shifter between I022 and the strip.

Wiring reference

Three wires, three connectors. This is the same for the bench test (step 13) and the final assembly (step 18).

BOARD PIGTAIL	STRIP WIRE	CARRIES	CONNECTOR
VIN (P1)	Red	5 V power	J1
GND (P1)	White (or black)	Ground	J2
IO22 (CN1)	Green	Data	J3



One joint per connector: a board wire in one port, the matching strip wire in another. The third port stays empty.

The two pigtails plug into P1 and CN1. You use three of their wires: VIN, GND and IO22. Tape and label them.

Red goes to VIN, never to 3V3. 3V3 can't supply the strip and you'll get a dim, flickering beam. **Green goes to IO22, not IO27** (they're neighbours on CN1). The unused wires on each pigtail (TX, RX, IO27, 3V3) get folded back and taped.

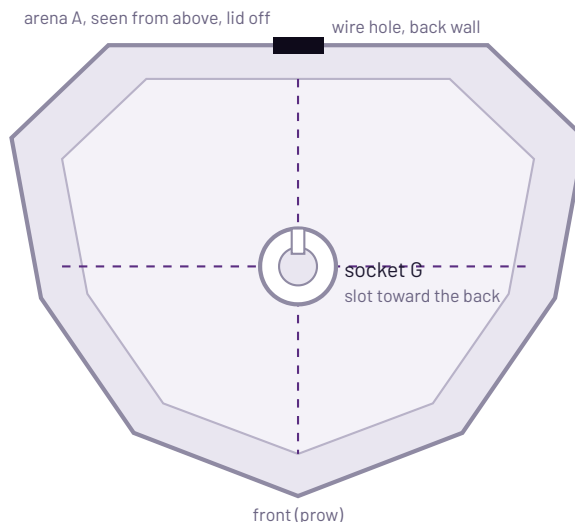
Mount the beam in the arena

14 Glue the socket to the arena floor

5 MIN + GLUE
TIME

1. Unplug the board and open the three lever nuts to free the strip. Leave the taped board leads in place.
2. Take arena base **A** and find the centre of the floor: measure the width and the depth of the inside floor with the ruler, mark the halfway point of each with a pencil, and mark where they cross.
3. Wipe the floor there and the underside of socket **G** with the alcohol wipe.
4. Socket **G** has a slot in its side. Sit it dry on the mark with the **slot facing the back** of the arena (the flat edge, where the wire hole is). Draw round it with the pencil.
5. Lift it, run a ring of hot glue on its underside (or four dots of super glue), and press it back onto the pencil outline. Hold for 30 seconds. Super glue: leave it 10 minutes.

Centre and level matter. The tube is 250 mm tall, so a socket 2 mm off-centre puts the tube off-centre in the lid hole, and a socket glued on a blob of glue makes the beam lean. Press it flat.

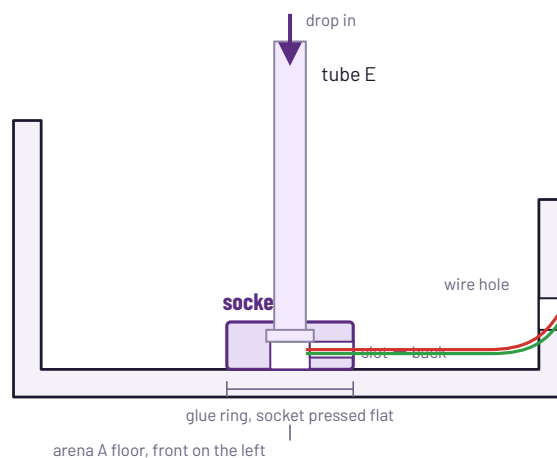


15 Stand the beam and route the wires

3
MIN

1. Pass the strip's three wires **through the socket's slot**, then along the floor and **out through the wire hole** in the back wall. Pull them through so there's no slack coiled inside.
2. Lower the tube's foot flange into the socket. It should drop in with a little play and stand straight up.
3. Sight down from the front and from the side: the tube should be vertical. If it leans, the socket isn't flat; reheat the glue and reset.

CHECK Tube stands straight, spins freely in the socket, wires exit the back of the arena with no strain on the strip.



Glue the socket dead centre with its slot toward the back. Wires go through the slot, along the floor and out the wire hole.

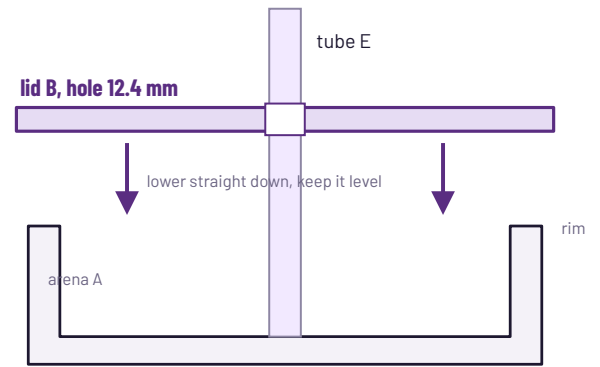
Fit the lid

16 Fit the lid

1 MIN

1. Hold lid **B** level above the arena, hole over the tube. Lower it straight down over the tube until it rests on the arena's rim.
2. Turn it until the roof's flat back edge lines up with the arena's flat back edge.

CHECK The lid sits flat all round and the tube passes through the hole without touching its edge. If it touches, the socket is off-centre (step 14) or the hole isn't centred (step 5).

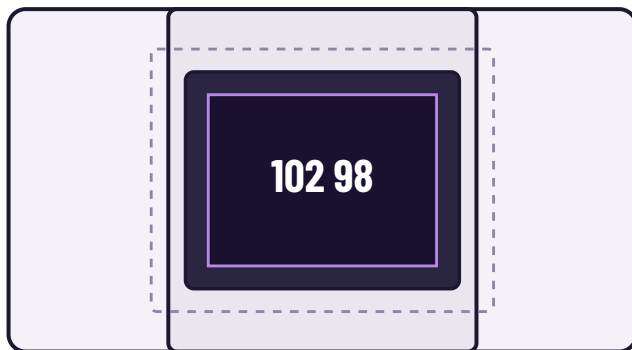


Keep the lid level as it goes on so the hole doesn't bind on the tube.

How the screen mounts

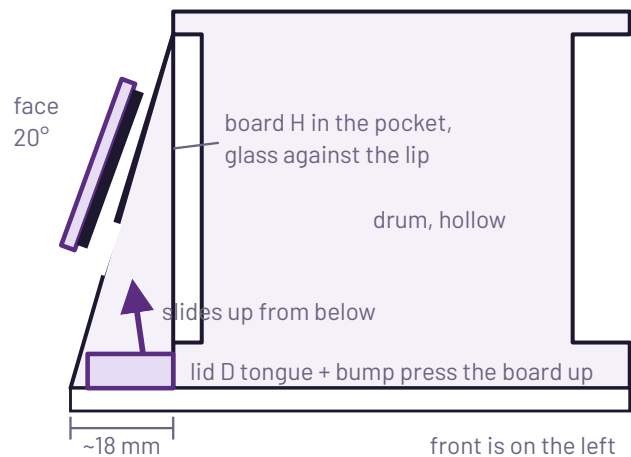
The screen lives in a small sloped **console** on the front of the plinth. Its face leans back 20° so the picture points up at you at a desk, and because the console stands in front of the drum, nothing above it blocks the view. Inside the console is a pocket, open toward the bottom: the board slides up into it, the window's lip stops the glass, and a tongue on the bottom lid presses the board up against that lip. No screws.

FRONT

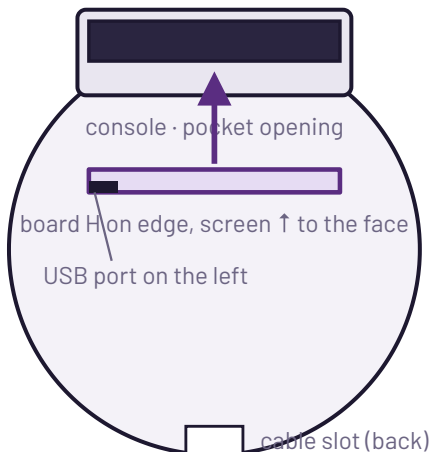


console 96 mm wide · window 60×46 · dashed: board inside
picture area centred in the window

SIDE, CUT THROUGH



BOTTOM, PLINTH UPSIDE DOWN



17 Seat the display board

5 MIN

1. Turn plinth **C** upside down, console toward you. The pocket opening is the rectangular slot in the underside of the console.
2. Unplug the two pigtails from board **H** so nothing snags. Hold the board on its long edge, **screen facing the sloped face, USB port on the left**.
3. Slide it into the pocket: light finger pressure, it stops when the glass meets the lip (same fit as step 4).
4. Plug the pigtails back into P1 and CN1; they hang out of the open bottom. Plug in the USB cable.
5. Turn the plinth right side up and look at the console: picture centred in the window, even black margin. Turn it back over for wiring.

Loose? A folded strip of masking tape on the back of the board (never the screen) takes up slack. The bottom lid's tongue and bump (step 19) also press the board up.

Connect the strip and close it up

18 Make the three connections (for real this time)

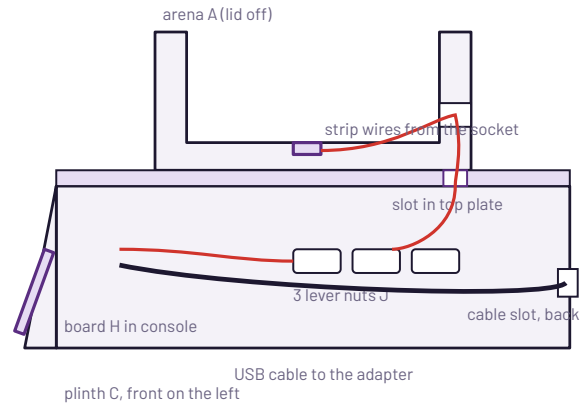
5
MIN

Same three joints as the bench test in step 13 (wiring reference on page 8), now with the strip inside the arena and the board inside the plinth.

BOARD PIGTAIL	STRIP WIRE	CARRIES	CONNECTOR
■ VIN (P1)	■ Red	5 V power	J1
■ GND (P1)	■ White (or black)	Ground	J2
■ IO22 (CN1)	■ Green	Data	J3

1. Set the arena on the table behind the plinth, with its wire hole facing the plinth's back.
2. Feed the strip's three wires down through the **slot in the plinth's top plate** at the back, into the plinth.
3. Through the open bottom of the plinth, make the three lever-nut joints per the table. Fold the thin board wires double as before. Close each lever firmly; you'll feel it click.
4. Tug-test all six wire ends.

Red goes to VIN, never to 3V3. 3V3 can't supply the strip and you'll get a dim, flickering beam. **Green goes to IO22, not IO27.**



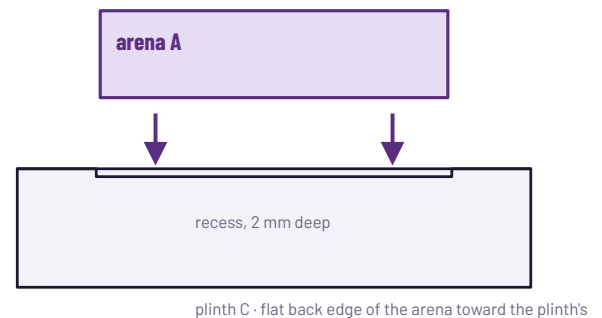
Strip wires: socket → wire hole → slot in the plinth top → lever nuts. Board USB cable → back slot.

19 Route the cable and close the plinth

5
MIN

1. Tuck the three connectors into the plinth, toward the back, away from the board.
2. Route the board's USB cable out the **slot at the back** of the plinth (or, if you fitted the optional panel jack, plug its short cable into the board and leave the jack's socket facing out).
3. Fit bottom lid **D**: its tongue goes **forward under the console**, finger hole toward the back. Press it in evenly until it's flush. The bump on the tongue lands under the board's bottom edge and pushes it up against the window lip.
4. Set the arena down onto the plinth. It drops into the shallow recess in the top plate; turn it so the flat back edge is at the back and the strip wires aren't pinched.

CHECK Plinth lid flush, arena seated with no rocking, one cable out the back. Looking in the front window, the screen is centred and straight.



Join WiFi and light the beam

20 Power on

1 MIN

1. Plug the USB cable into adapter **K** and the adapter into the wall.
2. The screen shows the Kings logo and **STARTING**, then **CONNECTING TO WIFI** for up to 20 seconds, then **WIFI SETUP**.

Use the **2 A adapter**, not a laptop port. The beam at full brightness plus the board draws about 1 A; a laptop port may brown out and restart the board mid-animation.



21 Connect it to your home WiFi from your phone

3 MIN

1. On your phone, open **Settings** → **WiFi**. A new network called **KingsBeam-Setup** appears (it can take 20 seconds). Join it. There's no password.
2. A setup page pops up on its own, like a hotel login page. If nothing appears within 15 seconds, open the browser and type **192.168.4.1** in the address bar.
3. Tap **Configure WiFi**. The page lists nearby networks. Tap **your home network**. If it isn't listed, tap Refresh; the board only sees 2.4 GHz networks, so pick the 2.4 GHz name if your router has separate ones.
4. Type your WiFi password carefully (it's case-sensitive; tap the eye icon to check it) and tap **Save**.
5. The setup network disappears; your phone reconnects to your normal WiFi by itself. The board's screen shows **CONNECTED** and then, within about 10 seconds, **NEXT GAME** with the opponent and tipoff time.

WIFI SETTINGS

KingsBeam-Setup >

Home-WiFi

Neighbour-5G

CoffeeShop

tap to join, no password

SETUP PAGE OPENS

KingsBeam-Setup

Configure WiFi

Info

no page? browse to
192.168.4.1

PICK YOUR NETWORK

Home-WiFi ■■■

Neighbour-5G ■■

Home-WiFi

.....

Save

2.4 GHz networks only

Join the setup network, tap Configure WiFi, choose your home network, type the password, Save.

Light the Beam

CHECK The **NEXT GAME** screen shows the right opponent and a sensible countdown. Time is synced from the internet, so the tipoff shown is already in Pacific time.

Using it

What the screen shows

NEXT GAME	Opponent logo, VS (home) or @ (away), tipoff in Pacific time, countdown pill. Refreshes every 10 minutes; every 30 seconds in the last 20 minutes before tipoff.
LIVE	Score, quarter and clock, refreshed every 20 seconds. Leader in white, trailer in grey with a marker under the leader. Halftime, end of quarter and overtime are labelled. The beam pulses softly.
FINAL, KINGS WIN	Score, "LIGHT THE BEAM", and the beam rises and holds.
FINAL, LOSS	Score and "NEXT TIME". The beam stays off unless you tap it.
NO GAME SCHEDULED	Off-season: the Kings logo and "NO GAME SCHEDULED".
OFFLINE	A small OFFLINE badge in the corner. The last good screen stays; it retries every minute.

Touch

TAP	Beam on / off, in any state.
HOLD 1 S	Cycle beam brightness: low → medium → full.

Automatic behaviour

AFTER A WIN	The beam stays lit up to 12 hours, until you tap it off, or until the next game shows up on the schedule.
AT NIGHT	Screen dims 11 pm–7 am, and whenever the room is dark (the board has a light sensor).
POWER LOSS	Reboots into the current state in ~15 seconds; WiFi is remembered.
NEW WIFI	If it can't find the saved network for 20 seconds after boot, KingsBeam-Setup opens again. Repeat step 21.

Care

Dust with a dry cloth. Keep out of direct sun and off radiators: PLA softens around 55 °C and the tube is thin. The lid lifts off to reach the beam; the bottom lid pulls off (finger hole) to reach the board. Unplug before opening.

Troubleshooting

SYMPTOM	TRY
Screen stays dark	Use the 2 A adapter, not a laptop. Try the board's other USB port if it has two. Check the cable is a data cable if you're flashing.
Colours inverted or milky	You have the other board variant. In <code>firmware/platformio.ini</code> , comment out <code>ILI9341_2_DRIVER</code> and enable the <code>ST7789_DRIVER + TFT_INVERSION_ON</code> lines noted there; reflash.
Picture upside down	In <code>firmware/src/ui.cpp</code> change <code>LV_DISPLAY_ROTATION_90</code> to <code>_270</code> ; reflash.
Screen off-centre in the window	Adjust <code>screen_ofs_x</code> / <code>screen_ofs_y</code> in <code>models/plinth.scad</code> by the offset in mm, <code>make</code> , reprint the test slice.
Beam never lights, even on tap	Unplug. Check red → VIN (not 3V3), green → IO22 (not IO27), white → GND. Check the strip's arrows point away from the wired end. Check each lever nut grips (tug test).
Only LED 1 lights / random colours	Data signal is marginal. Shorten the green lead to the minimum, or add the 74AHCT125 level shifter between IO22 and the strip's green wire.
Beam dim or flickers with the screen	Weak power. Use a 2 A adapter and a short USB cable. Step brightness down (hold 1 s) if it's a weak charger you can't change.
Beam leans	Socket not flat or off-centre: lift the lid, reheat the glue, reseal (step 14).
OFFLINE badge won't clear	ESPN unreachable from your network. It retries every minute. Check other devices have internet; reboot the board. If it persists for days, ESPN may have changed its edge rules; check <code>python3 tools/espn_probe.py</code> on your Mac.
Countdown wrong by hours	Clock not synced yet; wait a minute after connecting. If it stays wrong, your router blocks NTP (UDP 123).
Taps ignored	Press firmly with a fingertip in the middle of the screen; it's a resistive panel. Lower <code>Z_THRESHOLD</code> in <code>firmware/src/touch.cpp</code> if it never registers.
Setup network never appears	Power-cycle the board. It only opens the portal when it can't connect; if it already knows a WiFi and that WiFi is up, there's nothing to set up.

Specifications

FOOTPRINT	230 mm diameter (plinth)
HEIGHT	≈ 385 mm to the beam tip
POWER	5 V USB, ≈ 1 A peak, 0.3 A idle
NETWORK	2.4 GHz WiFi, outbound HTTPS only
DATA	ESPN public scores; no account, nothing stored
BOARD	ESP32-2432S028R, 2.8" 320 × 240 resistive touch
BEAM	15 × WS2812B in a 12 × 250 mm white PLA tube

Safety

Low-voltage USB only; never connect to anything but a 5 V USB adapter. Indoor use. Unplug before opening the plinth or touching the wiring. The beam tube is thin-walled and not a toy. Printed PLA parts are not food-safe and should be kept away from heat.

Files

Everything referenced here lives in the project repo: `firmware/` (build and flash), `models/` (OpenSCAD and STL exports), `docs/bom.md` (parts with links), `tools/` (helper scripts).